

Manuel Zechmann

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Software engineer with extensive experience in **C++** and **Python** across computer graphics, computer vision, and game development. Industry roles at Snap, Partium, and ViewApp.

EXPERIENCE

Snap Inc.

Vienna, Austria

Computer Vision Researcher

Aug 2025 – Apr 2026

- Built and operated large-scale synthetic data pipelines on Google Cloud (Vertex AI, Cloud Storage), generating and storing terabytes of data across distributed jobs.
- Researched 3D scene understanding and generative models: adapted network architectures and designed new loss objectives for novel use cases.
- Worked with Python, modern ML frameworks, VLMs, agentic workflows, and cloud-based experimentation.
- Collaborated with the team through reading groups and cross-functional discussions.

ViewApp

Vienna, Austria

Multimedia Software Engineer

Jun 2024 – Sep 2024, Jan 2025 – Jul 2025

- Developed C++ and Unreal Engine tools to streamline artists' workflows and improve production efficiency.
- Automated build processes using TeamCity, reducing manual work and deployment errors.
- Resolved engine-level bugs to enhance stability and user experience.

Partium

Vienna, Austria

Junior Software Engineer

Oct 2022 – Mar 2024

- Deployed and maintained Docker- and Kubernetes-based services in beta environments.
- Improved CI/CD pipelines and introduced developer tooling, increasing release reliability.
- Integrated OCR components and trained ML models for spare part recognition (Python).

Partium

Vienna, Austria

Software Engineer Intern

Jul 2022 – Sep 2022

- Evaluated and integrated Elasticsearch as a search backend, achieving faster retrieval on large datasets.
- Contributed to CI/CD and testing workflows within a cross-functional Agile team.

PROJECTS

VulkanParticles | Open-source C++ Vulkan engine

[GitHub](#)

PlanetGenerator | Open-source procedural planet generator; basis for GAMEON 2025 paper

[GitHub](#)

Tall Trunk Galaxy 64 | Super Mario 64 ROM hack written in C; SimpleFlips Slide Hack Competition

2024

EDUCATION

University of Vienna

Vienna, Austria

Master of Science (MSc), Media Computer Science, Grade Avg: 1.56

Mar 2023 – Nov 2025

- Master's thesis: Procedural Planet Generation with Level-of-Detail Optimization and Performance Analysis (Godot, GDScript). Presented at GAMEON 2025.
- Coursework in computer graphics, video game development, image synthesis, computer vision, and machine learning.

University of Vienna

Vienna, Austria

Bachelor of Science (BSc), Computer Science, Grade Avg: 2.32

Oct 2019 – Jan 2023

- Bachelor's thesis: Deep Learning-Based Hand Detection and Pose Recognition; procedural training dataset generator in Unity.
- Coursework in computer graphics, C++, and signal and image processing.

PUBLICATIONS

Comparative Analysis of Procedural Planet Generators

GAMEON 2025, Ghent

- Published and presented at the GAMEON 2025 Conference. Based on findings from my MSc thesis on procedural planet generation.

AWARDS

UNIVIE Research Award for Students

University of Vienna, 2025

- Awarded for publishing “Comparative Analysis of Procedural Planet Generators” at the GAMEON 2025 conference.

Leistungsstipendium (Merit Scholarship)

University of Vienna, 2026

- Received for outstanding academic performance.

SKILLS

Programming Languages: C++, Python, C, JavaScript, GDScript

Software Engineering: Docker, Kubernetes, CI/CD, Elasticsearch, Google Cloud (Vertex AI, GCS)

Domains: Computer Graphics, Computer Vision, Game Development, Machine Learning, Procedural Generation

Tools: Vulkan, Unreal Engine, PyTorch, TensorFlow, OpenGL, Godot, Git, TeamCity

Languages: German (Native), English (C2), Spanish (A2)